Banking Customer Churn Analysis Project Report

Project Title: Banking Customer Churn Analysis using Python & Tableau

Sector: Banking / Financial Services

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Team / Section: SectionE_G-12

Institute: Newton School of Technology

Academic Session: 2024–28

Project Type: Data Analytics & Business Intelligence

Executive Summary

Customer churn is a major concern in the banking sector because retaining an existing customer is significantly less expensive than acquiring a new one. Churn reduces revenue, impacts cross-selling opportunities, and weakens long-term customer lifetime value.

This project was conducted to analyze customer churn patterns using a structured banking dataset containing customer demographics, balances, tenure, transactions, and churn status. The project followed a complete analytics lifecycle including data cleaning, exploratory analysis, statistical testing, KPI development, and Tableau dashboard creation.

The study found that customers with lower balances, shorter relationship tenure, and lower engagement levels were more likely to churn. Around **18.53% of customers** in the dataset had churned, indicating moderate retention risk.

A Tableau dashboard was developed to help management monitor churn trends, segment customers, and take data-driven actions.

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1. Sector Context and Problem Statement

Banking Sector Context

The modern banking industry is highly competitive, with customers frequently switching banks due to:

- Better digital services
- Higher returns
- Better customer support
- Lower fees
- Personalized financial products

Since customer acquisition costs are high, banks prioritize retention.

Problem Statement

The bank needs to understand:

- Why customers churn
- Which customers are at highest risk
- Whether balance, tenure, age, or category impacts churn
- How to proactively reduce attrition

This project solves the problem through data analytics and dashboard intelligence.

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2. Data Description

Data Source

Structured customer banking dataset used for academic analytics purposes.

Dataset Structure

Metric	Value

Total Rows	28,382
Total Columns	20+
Format	CSV
Target Variable	Churn

Important Columns

Demographics

- age
- gender
- occupation
- dependents

Relationship

- vintage
- customer_tenure_years

Financial

- current_balance
- previous_month_balance
- current_month_credit
- current_month_debit

Segmentation

- customer_nw_category

Target

- churn (0 = retained, 1 = churned)

Dataset Limitations

- No income column

- No satisfaction score
- Limited geographic value
- Historical data only (not real-time)
- No complaint/service usage records

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3. Data Cleaning and ETL Methodology (Python Pipeline)

Performed using Python in Jupyter Notebook.

Extract

- Imported CSV using pandas
- Stored raw data in version-controlled repository

Transform

Missing Value Handling

Column	Treatment
gender	Mode
occupation	Mode
dependents	Median
city	Mode
last_transaction	Retained

Standardization

- Converted dates to datetime
- Fixed numeric data types
- Removed duplicates if any

Feature Engineering

Created:

- customer_tenure_years
- Balance Change
- Age Group
- Tenure Group
- Network Category Labels

Outlier Review

Extreme dependents values corrected.

Load

Final cleaned file exported:

`final_dashboard_dataset.csv`

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4. KPI and Metric Framework

KPI	Formula
Total Customers	Count(Customers)
Churned Customers	Sum(Churn)
Churn Rate %	Churned / Total
Avg Current Balance	Avg(Current Balance)

Avg Previous Balance	Avg(Previous Balance)
Avg Tenure	Avg(Customer Tenure Years)
Avg Age	Avg(Age)
Balance Growth	Current - Previous

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5. Exploratory Data Analysis (EDA)

Churn Distribution

- Retained: 81.47%
- Churned: 18.53%

Tenure Insights

Customers with longer tenure were more stable.

Balance Insights

Customers with lower balances churned more frequently.

Age Insights

Middle-aged customers represented a strong active customer base.

Category Insights

Higher net worth category customers maintained stronger balances.

Visualisations Used

- Histograms
- Boxplots
- Bar Charts
- Correlation Heatmap
- Balance Trend Charts

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6. Statistical Analysis Results

T-Test: Balance vs Churn

Metric	Value
T-statistic	5.576
P-value	2.51e-08

Result

Strong statistical evidence that balances differ between churned and retained customers.

Chi-Square Tests

Used for:

- Gender vs Churn
- Occupation vs Churn

Correlation Analysis

Important drivers:

- Balance
- Tenure
- Transaction activity

Logistic Regression

Used/planned to estimate churn probability.

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7. Dashboard Design (Tableau)

Dashboard Title

Banking Customer Segmentation & Retention Dashboard

Dashboard Layout

Top KPI Cards

- Total Customers
- Churn Rate
- Avg Balance
- Avg Age
- Avg Tenure

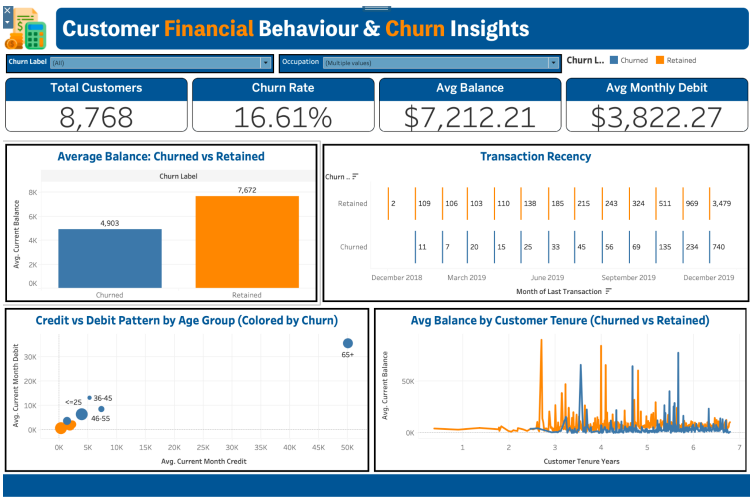
Main Charts

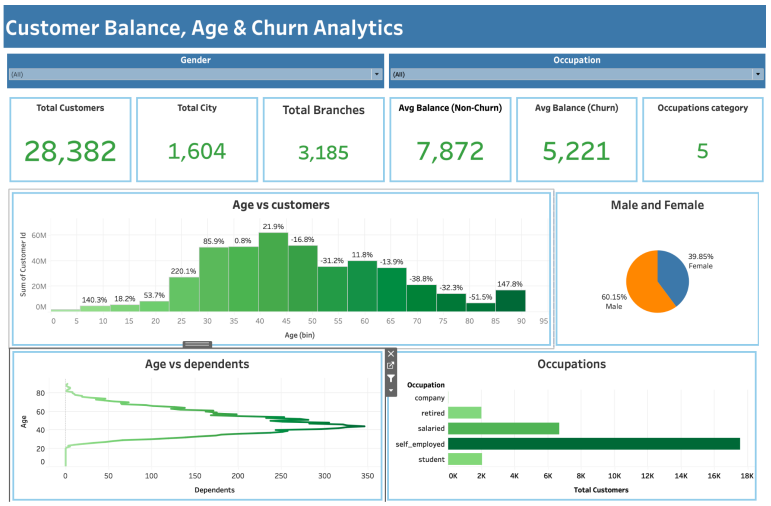
1. Avg Balance by Category
2. Current vs Previous Balance
3. Churn by Category
4. Churn by Tenure Group
5. Age vs Balance
6. Balance Growth by Category
7. Tenure Trend

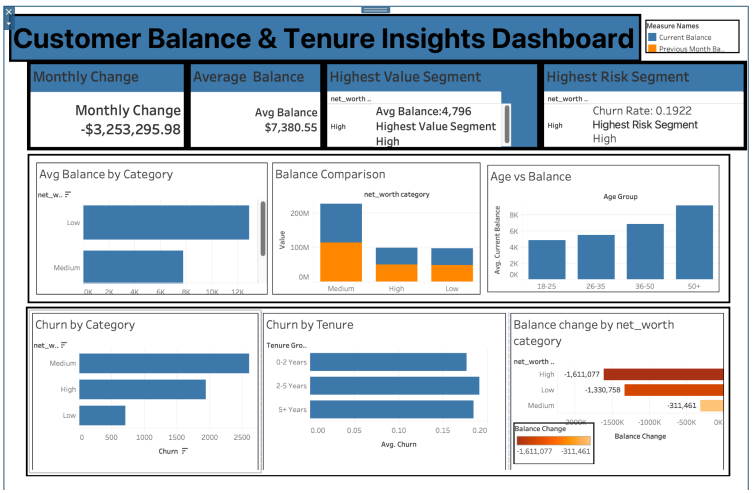
Filters

- Category
- Age Group
- Tenure Group
- Gender
- Occupation

Screenshot Placeholder







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8. Key Insights (Decision Language)

1. Approximately 1 in 5 customers churned, requiring immediate retention focus.
2. Low-balance customers are significantly more likely to leave.

3. Customers in first 2 years of relationship show highest churn risk.
4. High net worth customers hold disproportionate share of deposits.
5. Long-tenure customers are more loyal and financially stable.
6. Declining monthly balances may indicate early churn intent.
7. Age groups 26–50 contribute strong customer value.
8. Behavioral variables outperform demographic variables in churn prediction.
9. Premium customers should be proactively protected.
10. Customer segmentation can improve campaign ROI.
11. Churn risk can be monitored monthly through dashboard KPIs.
12. Retention strategies should differ by customer segment.

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9. Actionable Business Recommendations

1. Early Tenure Retention Program

Create onboarding journeys for new customers.

2. Low Balance Engagement Offers

Savings plans, bundled products, rewards.

3. Premium Customer Protection

Dedicated relationship management.

4. Automated Churn Alert System

Flag customers with declining balances.

5. Monthly Dashboard Review

Management should monitor churn KPIs regularly.

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10. Impact Estimation

If churn is reduced by even **3%**, the bank may achieve:

- Higher customer lifetime value
- Lower acquisition cost
- Increased deposits
- Better cross-sell conversion
- Improved profitability

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11. Limitations

- No real-time streaming data
- No customer satisfaction metrics
- No complaint history
- Limited regional intelligence
- No digital app usage data

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12. Future Scope

- Machine learning churn prediction
- Customer lifetime value model
- Real-time Power BI/Tableau dashboard
- Personalized offer engine
- Branch-wise retention model

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13. Contribution Matrix

Member	Contribution
Aditya Pal	ETL and Data Cleaning, EDA, Statistical Analysis, Tableau Dashboard, Final Report
Vansh Sharma	Dataset and Sourcing, Statistical Analysis, Tableau Dashboard, PPT & viva

Pratham Malhotra	Dataset and Sourcing, ELT and Cleaning, Tableau Dashboard
Sanal Nayyar	Tableau Dashboard, PPT & viva
Siddhant Giri	PPT & readme.md

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14. Tools & Technologies Used

Category	Tools
Programming	Python
Libraries	pandas, numpy, matplotlib, seaborn, scipy
Dashboarding	Tableau Public
Version Control	GitHub
Reporting	Jupyter Notebook

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15. Final Conclusion

This project successfully converted raw banking customer data into actionable business intelligence. The analysis identified balance, tenure, and engagement as major churn drivers. The dashboard now enables management to track churn risk, prioritize high-value customers, and take strategic retention actions.